

PROGRAMMING ASSIGNMENT RESULTS

WEEK 7

— Pointers, References, Memory, and C++ Essentials 1 Synthesis -
TRY 1

Course:	ESCC, CIS-251 - C++ Programming
Instructor:	Rosalyn Warren
Student:	Eric Hepperle

RUN 1: Normal list

3, then milk, bread, red apples on separate lines

```
ent\" ; if ($?) { g++ 102__Print-Grocery-List_Final.cpp -o 102_Pr
*****
*   Grocery List Program   *
*****

How many foods to enter in?: 3

Enter the next grocery item # 1: milk
Enter the next grocery item # 2: bread
Enter the next grocery item # 3: red apples
-----
* Successfully added 3 items! *
-----
***** GROCERY LIST *****
Item # 1: milk
Item # 2: bread
Item # 3: red apples
*****
```

RUN 2: Lower boundary

1, then eggs

```
*****
*   Grocery List Program   *
*****

How many foods to enter in?: 1

Enter the next grocery item # 1: eggs
-----
* Successfully added 1 items! *
-----
***** GROCERY LIST *****
Item # 1: eggs
*****
PS C:\Users\erich\Documents\tutwrk\ehw-coursework-escc\courses\Semest
```

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RUN 3: Upper boundary

10, then ten food names

```
*****
*   Grocery List Program   *
*****

How many foods to enter in?: 10

Enter the next grocery item # 1: apples
Enter the next grocery item # 2: oranges
Enter the next grocery item # 3: bananas
Enter the next grocery item # 4: kiwi
Enter the next grocery item # 5: nuts
Enter the next grocery item # 6: bread
Enter the next grocery item # 7: milk
Enter the next grocery item # 8: sugar
Enter the next grocery item # 9: coffee
Enter the next grocery item # 10: ice cream cones
-----
* Successfully added 10 items! *
-----
***** GROCERY LIST *****
Item # 1: apples
Item # 2: oranges
Item # 3: bananas
Item # 4: kiwi
Item # 5: nuts
Item # 6: bread
Item # 7: milk
Item # 8: sugar
Item # 9: coffee
Item # 10: ice cream cones
*****
PS C:\Users\erich\Documents\tutwrk\ehw-coursework-escc\courses\Semester_1_Fall_26\CIS251_Cpp_Progra
```

RUN 4: Below range

0

```
ent\" ; if ($?) { g++ 102__Print-Grocery-List_Final.cpp -o 102__Print-Grocery-List
*****
*   Grocery List Program   *
*****

How many foods to enter in?: 0
Invalid input: size must be between 1 and 10.
How many foods to enter in?: 200
Invalid input: size must be between 1 and 10.
How many foods to enter in?: -3
Invalid input: size must be between 1 and 10.
How many foods to enter in?: 2

Enter the next grocery item # 1: 
```

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RUN 5: Above range

11

```
ent\" ; if ($?) { g++ 102__Print-Grocery-List_Final.cpp -o 102__Print-Grocery-List_Final.exe
*****
*   Grocery List Program   *
*****

How many foods to enter in?: 11
Invalid input: size must be between 1 and 10.
How many foods to enter in?: -1
Invalid input: size must be between 1 and 10.
How many foods to enter in?: 2

Enter the next grocery item # 1: 
```

RUN 6: Non-numeric count

abc

```
ent\" ; if ($?) { g++ 102__Print-Grocery-List_Final.cpp -o 102__Print-Grocery-List_Final.exe
*****
*   Grocery List Program   *
*****

How many foods to enter in?: abc
Invalid input: size must be an integer (1-10)
Unable to get desired grocery count from user.
*****
```

RUN 7: Interrupted food input

Request 3; supply one food, then end input

```
ent\" ; if ($?) { g++ 102__Print-Grocery-List_Final.cpp -o 102__Print-Grocery-List_Final.exe
*****
*   Grocery List Program   *
*****

How many foods to enter in?: 3

Enter the next grocery item # 1: milk
Enter the next grocery item # 2: ^Z
Input failed.
Input stopped. Recorded 1 of 3 items.
***** GROCERY LIST *****
Item # 1: milk
*****
PS C:\Users\erich\Documents\tutwrk\ehw-coursework-escs\courses\Semester_1_Fall_26\CIS25
```

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RUN 8: Null-pointer guard

Temporarily call printGroceryItems(pFoods, 2) before allocation

```
int &itemsAddedCount);
void printGroceryItems(string *pFoods, int size);

int main()
{
    string *pFoods = nullptr;
    int size;          // requested number of
    array elements to allocate
    int itemsAddedCount = 0; // number of
    successfully entered foods

    //Testing: Temporarily call printGroceryList
    (pFoods, 0) before allocation
    printGroceryItems(pFoods, 2);
}
```

```
PS C:\Users\erich\Documents\_tutwrk\ehw-courses\Semester_1_Fall_26\CIS251_Cpp_Proj\grocery-List_Final>
Sorry, no grocery list is available.

*****
*   Grocery List Program   *
*****

How many foods to enter in?:
```

RUN 9: Items added is zero

Temporarily set itemsAddedCount to 0 inside printGroceryItems(), 1, then nuts

```
175
176 // Inspect the grocery items through a pointer without changing
177 them.
178 void printGroceryItems(string *pFoods, int itemsAddedCount)
179 {
180     // Temp Testing: change items added to zero
181     itemsAddedCount = 0;
182
183     // Check for nullptr before dereferencing pFoods
184     if (pFoods == nullptr)
185     {
186         cout << "Retrieving grocery list...\n";
187         cout << "Sorry, no grocery list was found.\n\n";
188         return;
189     }
190
191     // Print the grocery list
192     for (int i = 0; i < itemsAddedCount; i++)
193     {
194         cout << pFoods[i] << "\n";
195     }
196
197     cout << "Successfully added " << itemsAddedCount << " items!\n";
198 }
```

```
oursework-estc\courses\semester_1_fall_26\grocery-List_Final> .\102_Print-Grocery-List
*****
*   Grocery List Program   *
*****

How many foods to enter in?: 1

Enter the next grocery item # 1: nuts
-----
* Successfully added 1 items! *
-----

Sorry, no items were found!
*****
PS C:\Users\erich\Documents\_tutwrk\ehw-c
```